

securities you already own.” But the way they trade—dynamically churning securities, employing leverage and short positions, and trading derivatives—allows HFs to emphasize some risk factors that play minor roles in long-only, passive positions of “securities you already own.” Before we measure their risk exposures (or betas), a note of caution is in order about what constitutes HF returns.

3.1. DATA BIASES

Dead men tell no tales. *Survivorship bias* arises because HFs voluntarily report to public databases. HFs don’t like to report if their returns are bad—so they stop reporting when they are in distress. Therefore, we tend not to see bankrupt or liquidated funds in the databases because they remove themselves before they die. This causes an upward bias in reported HF returns. Another type of survivorship bias is caused by successful funds retroactively reporting past returns to the databases (a process called *backfilling*)—again you tend to provide this information only if the fund is doing well and past returns are good.

However, HFs report to databases to advertise themselves. (HFs can’t solicit money through public offerings). If you don’t need to attract investors, why report? It’s better to remain secretive. As the best HFs often are not found in the databases, there is also a downward bias in reported returns.

Thus, HF databases miss the extreme left- and right-hand tails. Which effect dominates? Researchers have concluded that there are many more bad HFs that stop reporting than good HFs that never report. (Technically, the databases truncate the left-hand tail.) HF databases, therefore, reflect returns that are too good.

How large is the survivorship bias? At least 2%, and most researchers peg the bias at 3% to 4% or more.²³ Malkiel and Saha (2005) estimate the return of HFs that backfill is 14.7% compared to 7.3% for funds that do not backfill. This is a very large backfill bias of 7.3%. They estimate the survivorship bias to be 4.4%, which is the difference in the average return of live HFs compared to the returns of HFs that stop reporting or die. Fung, Xu, and Yau (2004) argue that the survivorship bias is large enough to completely wipe out any outperformance of HFs in reported data compared to boring stocks and bonds, if outperformance even exists in the first place.

Aiken, Clifford, and Ellis (2013) cleverly estimate the survivorship bias by finding a sample of HFs that do not report to the main hedge fund databases. They can see what happens to HFs that originally report and then stop. They conclude that “commercial databases are missing the worst performers in the hedge fund universe.” The difference in risk-adjusted returns between reporting HFs and nonreporting HFs is 3.5% using size, value-growth, and momentum factors (see

²³ In addition to the references in the text, see also Fung and Hsieh (1997), Brown, Goetzmann, and Ibbotson (1999), Liang (2000), and Agarwal, Fos, and Jiang (2013).